In [1845]:

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount
("/content/drive", force_remount=True).
```

In [1846]:

```
import pandas as pd
from matplotlib import pyplot as plt
import seaborn as sns
import matplotlib.pyplot as plt
import matplotlib.ticker as ticker
from scipy import stats
```

In [1847]:

```
#data=pd.read_csv('/content/drive/MyDrive/Colab Notebooks/STIC/Neo4j/TIEMPOS CONSULTAS Y
CONSUMO ENERGÉTICO - CSV 10GB.csv')
#data=pd.read_csv('/content/drive/MyDrive/Colab Notebooks/STIC/Neo4j/TIEMPOS CONSULTAS Y
CONSUMO ENERGÉTICO - CSV 30GB.csv')
data=pd.read_csv('/content/drive/MyDrive/Colab Notebooks/CSV 10GB.csv')
#data=pd.read_csv('/content/drive/MyDrive/Colab Notebooks/STIC/Neo4j/CSV 30GB.csv')

data
```

Out[1847]:

	Query	Iteraci�	tiempoSD	tiempoCD	energySD	energyCD
0	1	1	38.501	36.445	50.7523	129.8160
1	1	2	28.212	18.050	24.7744	17.0540
2	1	3	28.100	16.755	23.8491	15.2390
3	1	4	29.289	15.605	24.7466	13.2440
4	1	5	27.385	17.678	23.0229	15.3716
...	...	...	...	...	...	...
245	25	6	31.597	0.311	26.1	0.4220
246	25	7	31.523	0.311	25.665	0.5040
247	25	8	31.393	0.341	25.534	0.4600
248	25	9	32.229	0.318	26.175	0.4920
249	25	10	32.849	0.366	30.52	1.8360

250 rows × 6 columns

In [1847]:

```
```

In [1848]:

```
nulos_por_columna = data.isnull().sum()

print(nulos_por_columna)
```

```
Query          0
Iteraci�       0
tiempoSD       0
tiempoCD      10
energySD       0
energyCD      10
```

```
dtype: int64
```

In [1849]:

```
data.columns=['Query', 'Iteración', 'tiempoSD', 'tiempoCD', 'energySD','energyCD']
```

In [1850]:

```
def quitar_commas(x):
    return float(str(x).replace(',', ''))

data['tiempoSD'] = data['tiempoSD'].apply(quitar_commas)
data['tiempoCD'] = data['tiempoCD'].apply(quitar_commas)
data['energySD'] = data['energySD'].apply(quitar_commas)
data['energyCD'] = data['energyCD'].apply(quitar_commas)
```

In [1851]:

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 250 entries, 0 to 249
Data columns (total 6 columns):
 #   Column     Non-Null Count  Dtype  
---  ------     --------------  -----  
 0   Query      250 non-null    int64  
 1   Iteración  250 non-null    int64  
 2   tiempoSD   250 non-null    float64
 3   tiempoCD   240 non-null    float64
 4   energySD   250 non-null    float64
 5   energyCD   240 non-null    float64
dtypes: float64(4), int64(2)
memory usage: 11.8 KB
```

In [1852]:

```
data.describe()
```

Out[1852]:

	Query	Iteración	tiempoSD	tiempoCD	energySD	energyCD
count	250.000000	250.000000	250.000000	240.000000	250.000000	240.000000
mean	13.000000	5.500000	888.306276	222.036900	419.400285	171.058330
std	7.225568	2.878043	2519.678888	923.985083	1052.439558	644.729441
min	1.000000	1.000000	1.066000	0.111000	1.001420	0.123000
25%	7.000000	3.000000	2.732000	0.660500	3.708592	0.977250
50%	13.000000	5.500000	14.067500	7.344000	13.566650	6.711800
75%	19.000000	8.000000	38.384000	32.646525	45.776375	32.366750
max	25.000000	10.000000	11081.073000	4634.875000	3378.770000	3238.680000

In [1853]:

```
data.shape
```

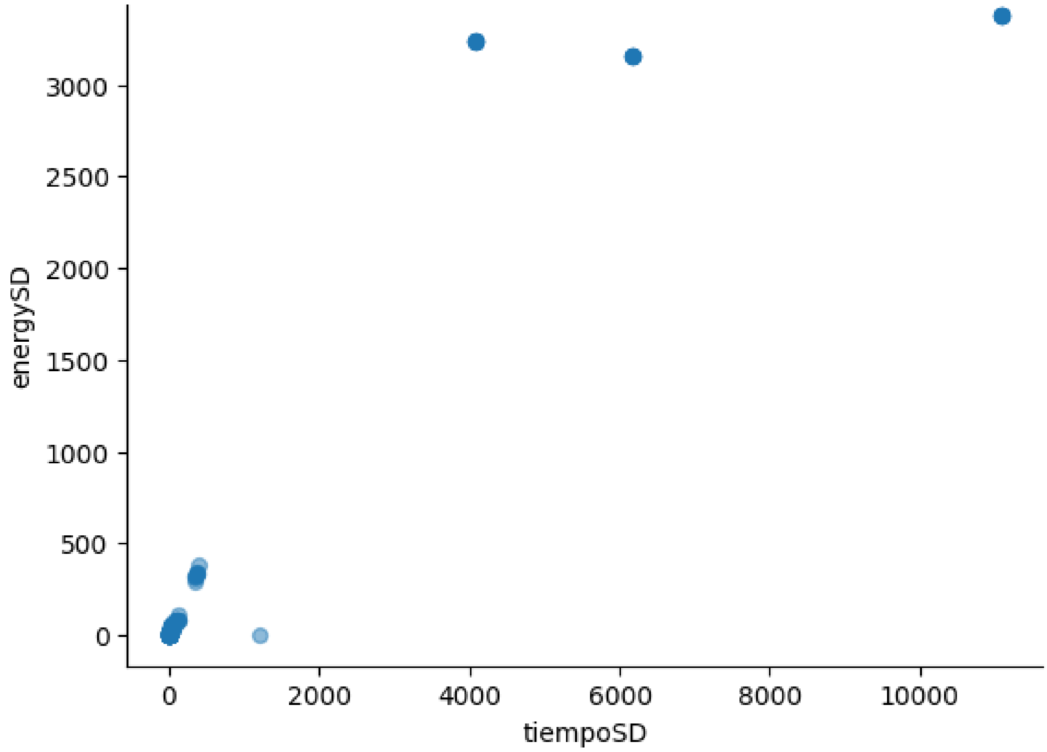
Out[1853]:

```
(250, 6)
```

In [1854]:

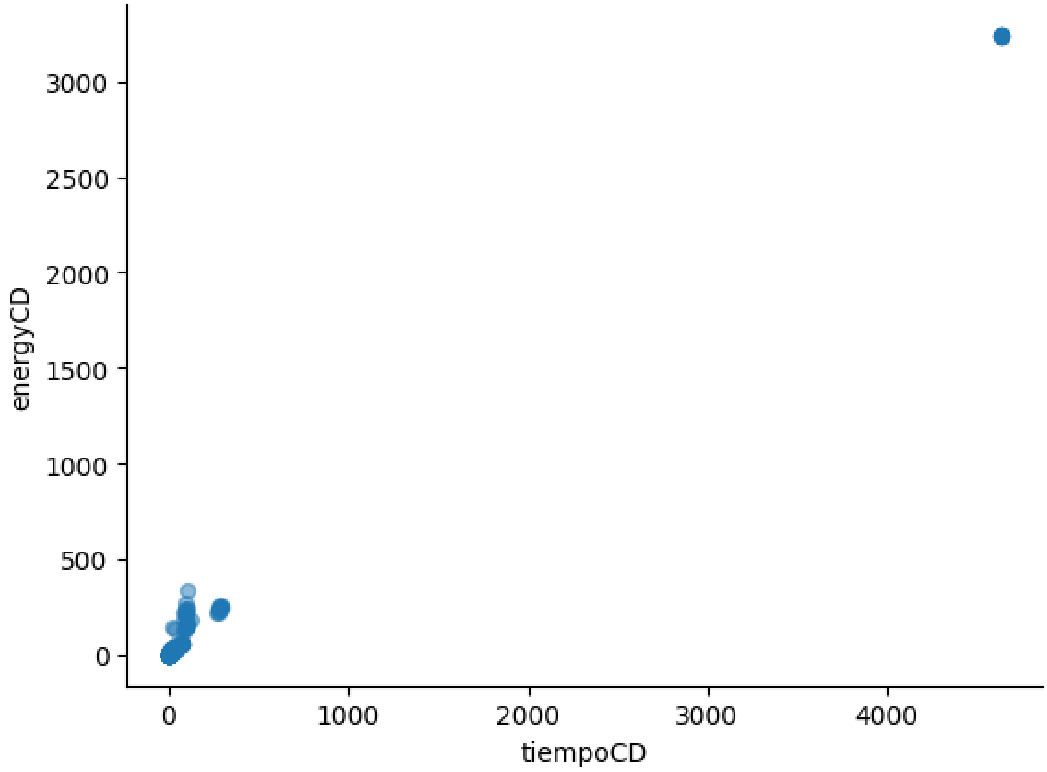
```
# @title Tiempo vs Consumo (sin diseño)

data.plot(kind='scatter', x='tiempoSD', y='energySD', s=32, alpha=.5)
plt.gca().spines[['top', 'right',]].set_visible(False)
```

3500



In [1855]:

```
# @title Tiempo vs Consumo (con diseño)

data.plot(kind='scatter', x='tiempoCD', y='energyCD', s=32, alpha=.5)
plt.gca().spines[['top', 'right',]].set_visible(False)
```



In [1856]:

```
#data.drop(data[data['tiempoSD'] > 1500].index, inplace=True)
```

In [1857]:

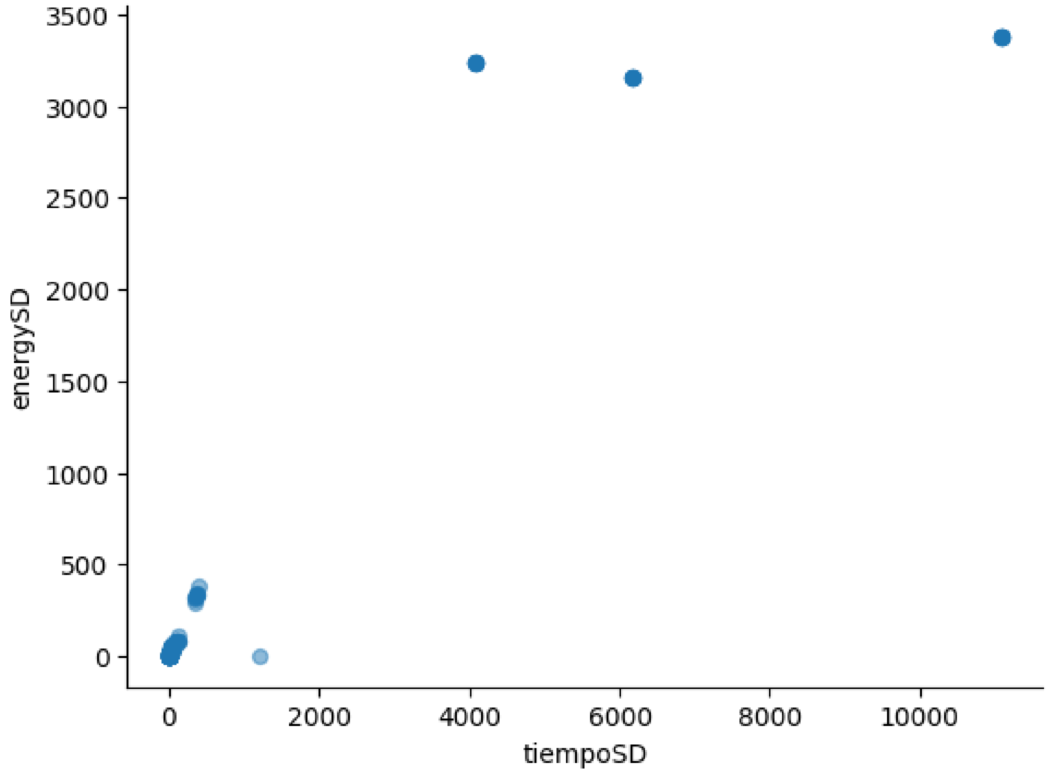
```
data.shape
```

Out[1857]:

```
(250, 6)
```

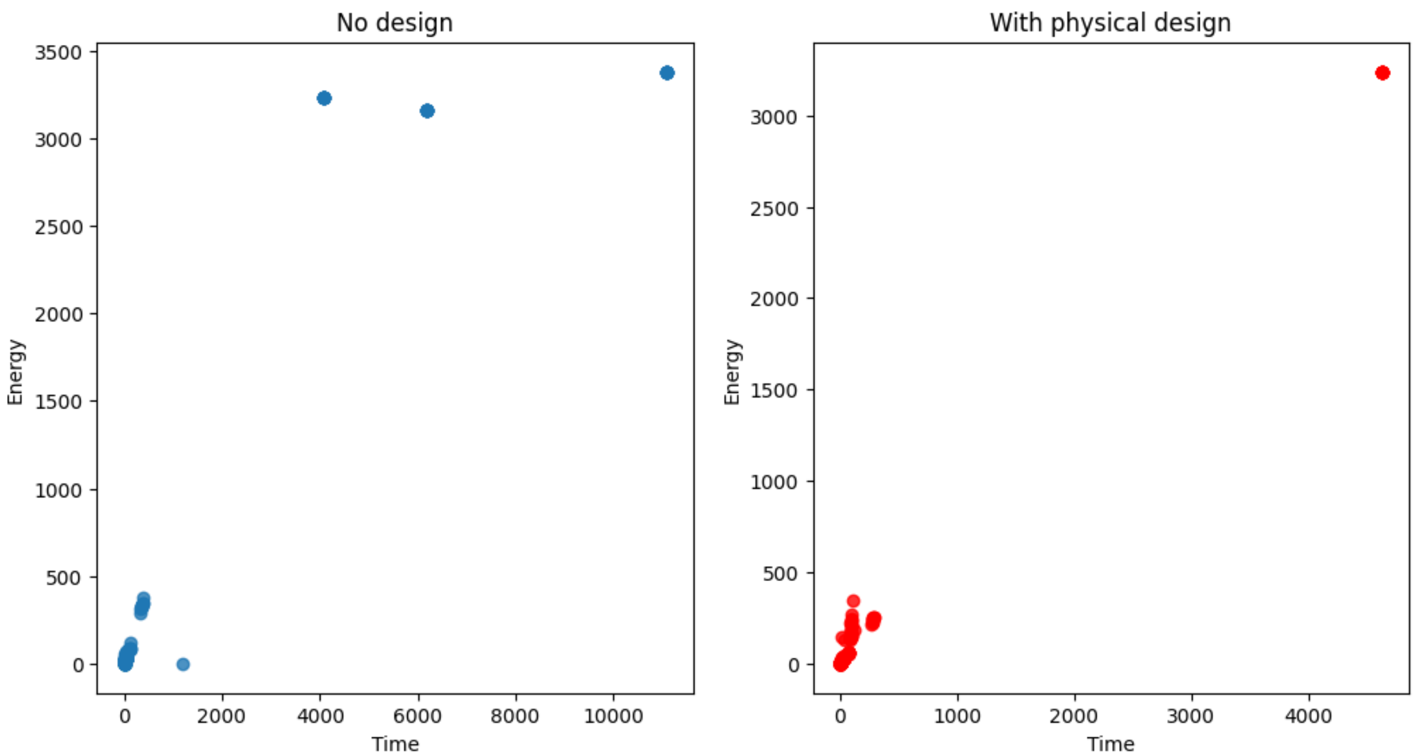
In [1858]:

```
data.plot(kind='scatter', x='tiempoSD', y='energySD', s=32, alpha=.5)
plt.gca().spines[['top', 'right',]].set_visible(False)
```



In [1859]:

```
fig, ax = plt.subplots(1, 2, figsize=(12,6))
ax[0].scatter(x=data.tiempoSD, y=data.energySD, alpha= 0.8)
ax[0].set_title('No design')
ax[0].set_xlabel('Time')
ax[0].set_ylabel('Energy');
ax[1].scatter(x=data.tiempoCD, y=data.energyCD, alpha= 0.8,color='red')
ax[1].set_title('With physical design')
ax[1].set_xlabel('Time')
ax[1].set_ylabel('Energy');
fig.savefig("Correlation10GB.jpg", bbox_inches='tight', dpi=300)
```

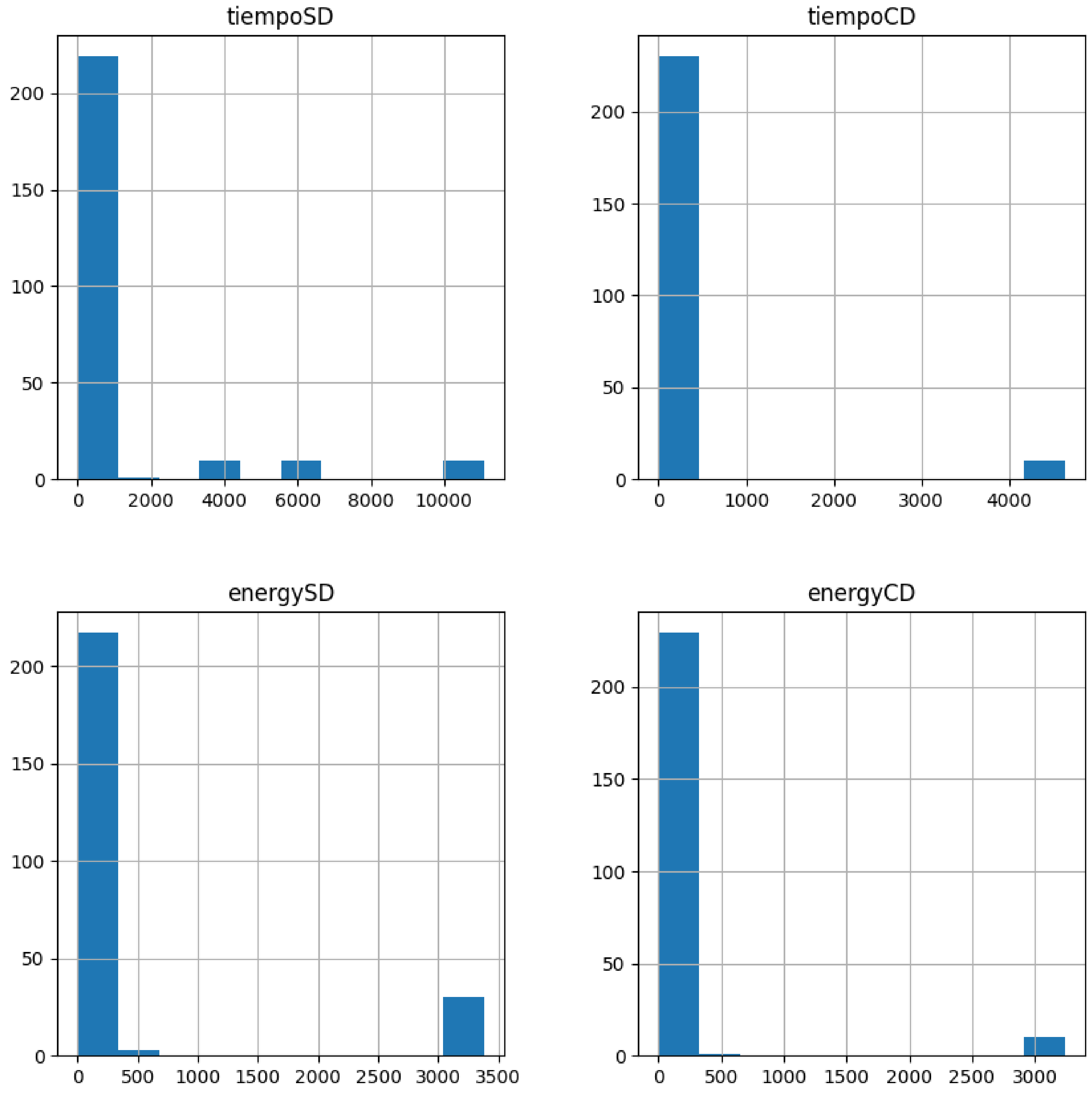


In [1860]:

```
data[['tiempoSD','tiempoCD','energySD','energyCD']].hist(figsize=(10,10))
```

Out[1860]:

```
array([[<Axes: title={'center': 'tiempoSD'}>,
        <Axes: title={'center': 'tiempoCD'}>],
       [<Axes: title={'center': 'energySD'}>,
        <Axes: title={'center': 'energyCD'}>]], dtype=object)
```



In [1861]:

```
data_sinnulos = data.dropna()
data_sinnulos.shape
```

Out[1861]:

```
(240, 6)
```

In [1862]:

```
# Normalidad de los residuos Shapiro-Wilk test
# ==============================================================================
shapiro_test = stats.shapiro(data_sinnulos.tiempoSD)
print(f"Variable height: {shapiro_test}")
shapiro_test = stats.shapiro(data_sinnulos.tiempoCD)
```

```
print(f"Variable weight: {shapiro_test}")
shapiro_test = stats.shapiro(data_sinnulos.energySD)
print(f"Variable height: {shapiro_test}")
shapiro_test = stats.shapiro(data_sinnulos.energyCD)
print(f"Variable weight: {shapiro_test}")
```

```
Variable height: ShapiroResult(statistic=np.float64(0.40060305004698704), pvalue=np.float
64(1.6577028181157293e-27))
Variable weight: ShapiroResult(statistic=np.float64(0.2335306833488937), pvalue=np.float6
4(2.806452324111487e-30))
Variable height: ShapiroResult(statistic=np.float64(0.423464969107268 7), pvalue=np.float6
4(4.415440201649857e-27))
Variable weight: ShapiroResult(statistic=np.float64(0.2612306637149038), pvalue=np.float6
4(7.447185136002885e-30))
```

In [1863]:

```
data.describe()
```

Out[1863]:

	Query	Iteración	tiempoSD	tiempoCD	energySD	energyCD
count	250.000000	250.000000	250.000000	240.000000	250.000000	240.000000
mean	13.000000	5.500000	888.306276	222.036900	419.400285	171.058330
std	7.225568	2.878043	2519.678888	923.985083	1052.439558	644.729441
min	1.000000	1.000000	1.066000	0.111000	1.001420	0.123000
25%	7.000000	3.000000	2.732000	0.660500	3.708592	0.977250
50%	13.000000	5.500000	14.067500	7.344000	13.566650	6.711800
75%	19.000000	8.000000	38.384000	32.646525	45.776375	32.366750
max	25.000000	10.000000	11081.073000	4634.875000	3378.770000	3238.680000

In [1864]:

```
# Cálculo de correlación y significancia
# ===================================================================================

print('Correlación Pearson: ', data['tiempoSD'].corr(data['energySD'], method='pearson')
)
print('Correlación spearman: ', data['tiempoSD'].corr(data['energySD'], method='spearman
'))
print('Correlación kendall: ', data['tiempoSD'].corr(data['energySD'], method='kendall',
))
```

```
Correlación Pearson:  0.9229497783305903
Correlación spearman:  0.9673087064248347
Correlación kendall:  0.897221595226579
```

In [1864]:

```

```

In [1865]:

```
# Cálculo de correlación y significancia con Scipy
# ===================================================================================
from scipy import stats
r, p = stats.pearsonr(data_sinnulos['tiempoSD'], data_sinnulos['energySD'])
print(f"Correlación Pearson: r={r}, p-value={p}")

r, p = stats.spearmanr(data_sinnulos['tiempoSD'], data_sinnulos['energySD'])
print(f"Correlación Spearman: r={r}, p-value={p}")

r, p = stats.kendalltau(data_sinnulos['tiempoSD'], data_sinnulos['energySD'])
print(f"Correlación Pearson: r={r}, p-value={p}")
```

```
Correlación Pearson: r=0.9231050314591467, p-value=9.243601242639186e-101
```

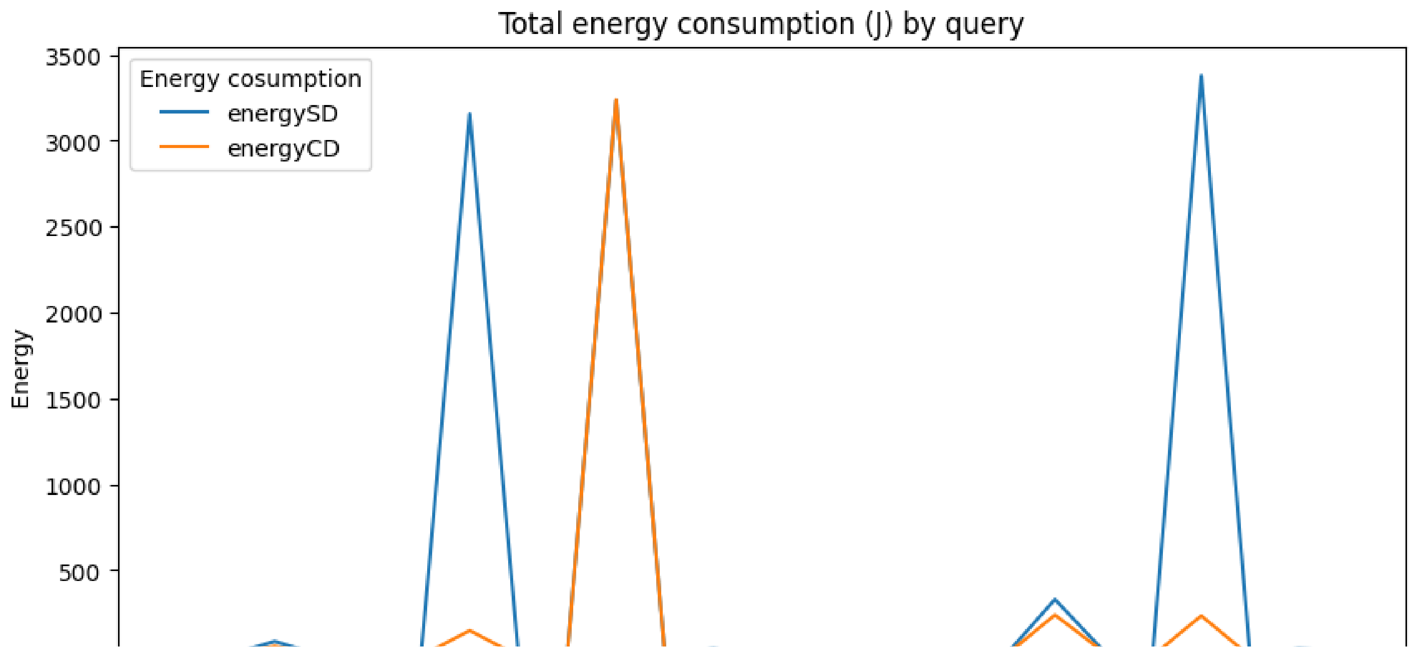
```
Correlación Spearman: r=0.987564815425904, p-value=3.036929537589074e-193
Correlación Pearson: r=0.9156740476471303, p-value=3.548019362469 97e-98
```

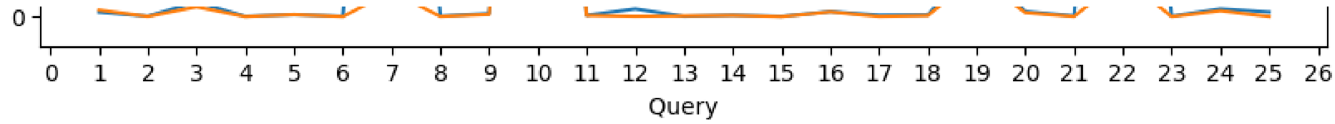
In [1866]:

```python
fig, ax = plt.subplots(figsize = (10, 5))
ax.xaxis.set_major_locator(ticker.MultipleLocator(1))
ax.set (xlabel = ' Query ', ylabel = ' Time ',  title = ' Total Time Execution (s) by qu
ery ')
resumen_total_time=data[['Query','tiempoSD','tiempoCD']]
sns.lineplot(x='Query', y='value', hue='variable', data=pd.melt(resumen_total_time, ['Qu
ery']), errorbar=None)
legend = plt.legend( title="Time execution")
```

In [1867]:

```python
fig, ax = plt.subplots(figsize = (10, 5))
ax.xaxis.set_major_locator(ticker.MultipleLocator(1))
ax.set (xlabel = ' Query ', ylabel = ' Energy ',  title = ' Total energy consumption (J)
by query ')
resumen_total_energy=data[['Query','energySD','energyCD']]
sns.lineplot(x='Query', y='value', hue='variable', data=pd.melt(resumen_total_energy, ['
Query']), errorbar=None)
legend = plt.legend( title="Energy cosumption")
```

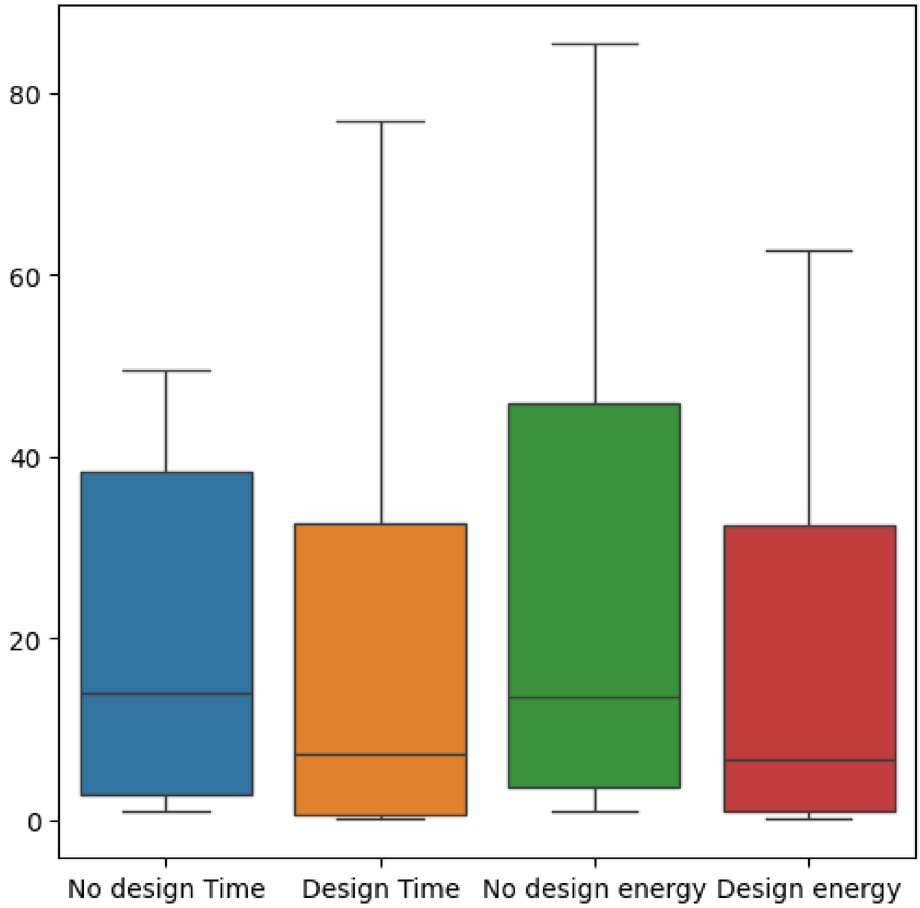




In [1868]:

```
fig, ax = plt.subplots(nrows=1, ncols=1, figsize=(6, 6))
sns.boxplot(data=data.loc[:, ['tiempoSD', 'tiempoCD', 'energySD','energyCD']],showfliers=
False)
ax.set_xticklabels(['No design Time', 'Design Time', 'No design energy','Design energy']
)
plt.savefig("Bloxpot10GB.jpg", bbox_inches='tight', dpi=300)
plt.show()
```

```
<ipython-input-1868-42d34e36e68e>:3: UserWarning: set_ticklabels() should only be used wi
th a fixed number of ticks, i.e. after set_ticks() or using a FixedLocator.
  ax.set_xticklabels(['No design Time', 'Design Time', 'No design energy','Design energy'
])
```

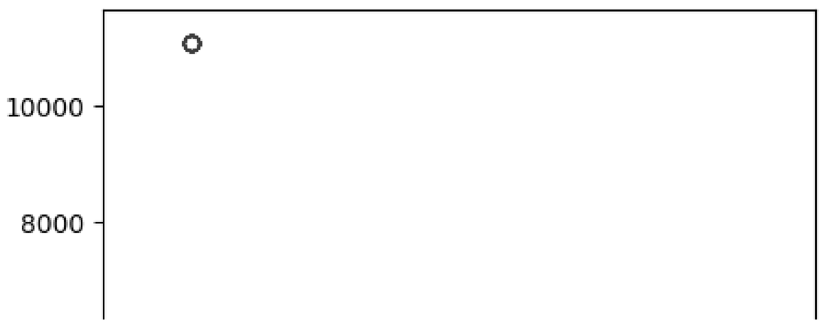


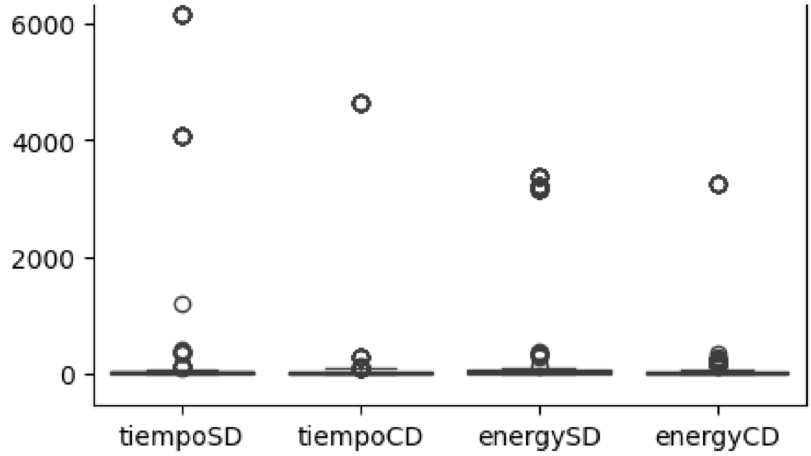
In [1869]:

```
fig, ax = plt.subplots(nrows=1, ncols=1, figsize=(5, 5))
#sns.boxplot(data=data.loc[:, ['tiempoSD', 'tiempoCD', 'energySD','energyCD']],showfliers
=False)
sns.boxplot(data=data.loc[:, ['tiempoSD', 'tiempoCD', 'energySD','energyCD']])
```

Out[1869]:

<Axes: >





In [1870]:

```
grouped_mean=data[['Query','tiempoSD', 'tiempoCD', 'energySD', 'energyCD']].groupby("Query").mean().round(2)
grouped_mean
```

Out[1870]:

Query	tiempoSD	tiempoCD	energySD	energyCD
1	28.62	18.77	26.19	39.22
2	2.39	0.79	3.30	1.44
3	112.83	74.00	86.01	57.95
4	1.82	0.15	2.18	0.59
5	13.95	14.36	12.21	12.47
6	2.16	0.41	2.43	0.62
7	6174.36	95.45	3157.77	149.03
8	2.79	0.80	3.71	1.16
9	18.71	14.02	18.39	13.86
10	4074.47	4634.88	3234.78	3238.68
11	9.59	8.36	8.35	7.40
12	36.19	1.75	46.03	2.13
13	121.19	NaN	2.21	NaN
14	8.39	7.51	8.80	7.54
15	1.78	0.16	1.94	0.43
16	32.34	30.24	29.69	28.31
17	9.44	0.86	9.16	1.28
18	10.72	6.76	10.59	6.77
19	364.81	280.86	330.55	240.08
20	19.63	8.44	30.52	23.82
21	3.58	0.46	4.72	1.04
22	11081.07	94.32	3378.77	235.17
23	2.32	0.31	3.12	0.55
24	42.24	34.89	46.07	34.68
25	32.25	0.36	27.52	1.17

In [1871]:

```
grouped_std=data[['Query','tiempoSD', 'tiempoCD', 'energySD', 'energyCD']].groupby("Quer
```

```
y").std().round(2)
grouped_std
```

Out[1871]:

Query	tiempoSD	tiempoCD	energySD	energyCD
1	3.58	6.27	8.67	52.18
2	0.32	0.19	2.03	1.55
3	8.71	2.79	10.63	2.40
4	0.27	0.05	0.90	0.42
5	0.75	0.87	1.99	1.72
6	0.31	0.10	0.94	0.44
7	0.00	9.22	0.00	16.54
8	0.57	0.08	2.94	0.47
9	0.89	1.65	1.67	3.12
10	0.00	0.00	0.00	0.00
11	5.38	4.53	3.98	4.08
12	3.79	1.11	5.94	1.17
13	379.06	NaN	2.07	NaN
14	1.17	0.38	2.70	1.46
15	0.15	0.06	0.63	0.26
16	2.29	5.03	1.92	4.75
17	0.60	0.14	1.44	0.73
18	0.34	0.47	1.58	2.11
19	18.68	5.54	23.18	12.71
20	0.61	0.48	8.83	4.78
21	0.33	0.07	2.30	0.53
22	0.00	5.91	0.00	44.60
23	0.44	0.04	2.44	0.21
24	2.97	2.55	10.12	4.42
25	0.82	0.09	2.24	0.90

In [1872]:

```
tabla=pd.merge(left=grouped_mean,right=grouped_std, left_index=True, right_index=True)
```

In [1873]:

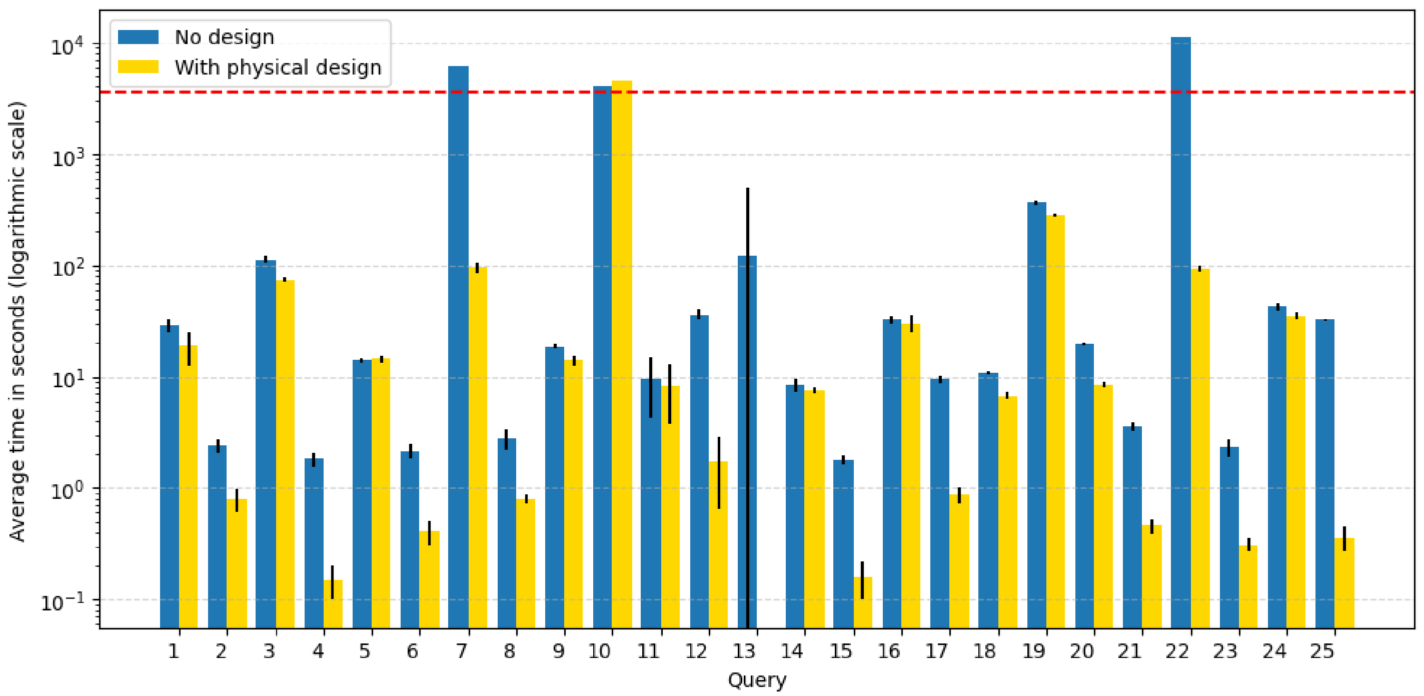
```
tabla.head()
```

Out[1873]:

Query	tiempoSD_x	tiempoCD_x	energySD_x	energyCD_x	tiempoSD_y	tiempoCD_y	energySD_y	energyCD_y
1	28.62	18.77	26.19	39.22	3.58	6.27	8.67	52.18
2	2.39	0.79	3.30	1.44	0.32	0.19	2.03	1.55
3	112.83	74.00	86.01	57.95	8.71	2.79	10.63	2.40
4	1.82	0.15	2.18	0.59	0.27	0.05	0.90	0.42
5	13.95	14.36	12.21	12.47	0.75	0.87	1.99	1.72

In [1874]:

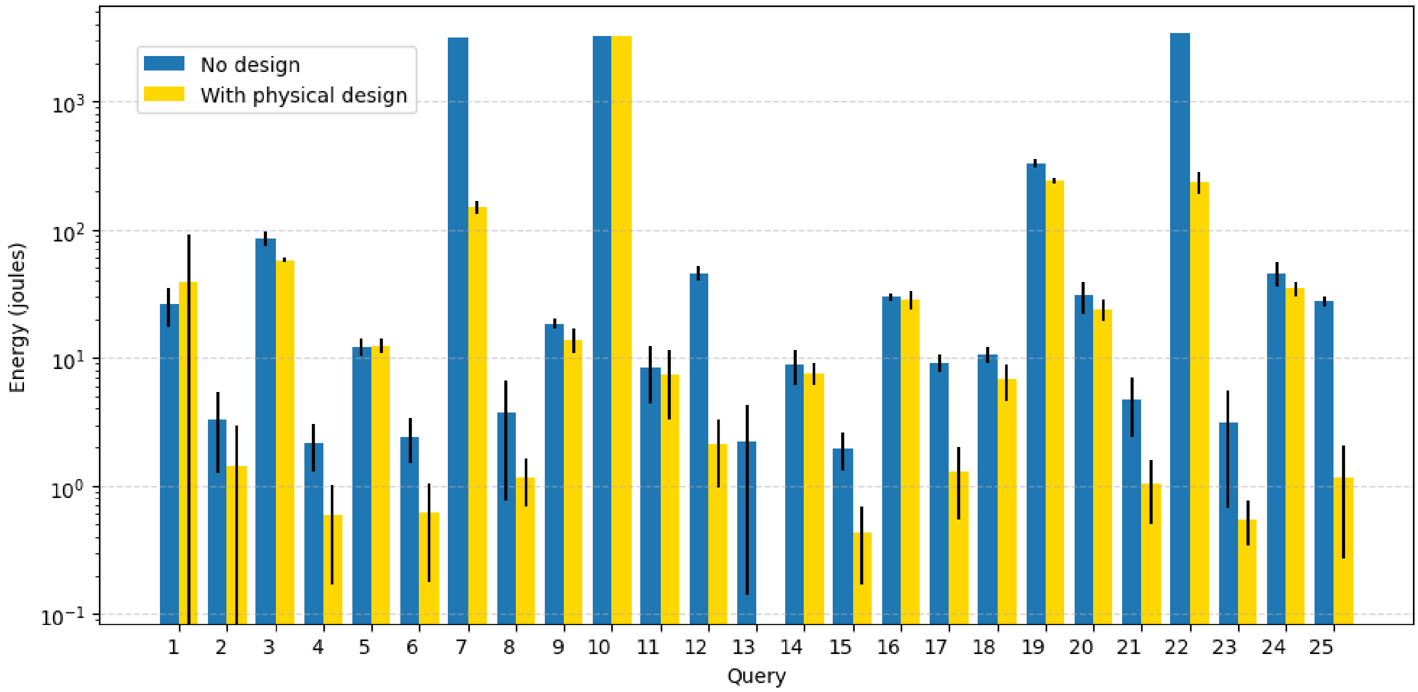
```
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files # Import files module

x = np.arange(len(tabla))   # posición de cada grupo
bar_width = 0.4

# Crear gráfico
fig, ax = plt.subplots(figsize=(10, 5))

# Colores definidos
color_validate = '#1f77b4'   # azul
#color_query = '#ff7f0e'      # naranja

color_query = '#FFD700'      # naranja

# Dibujar barras
ax.bar(x - bar_width/2, tabla['tiempoSD_x'], width=bar_width, label='No design', yerr=ta
bla['tiempoSD_y'], color=color_validate)
ax.bar(x + bar_width/2, tabla['tiempoCD_x'], width=bar_width, label='With physical desig
n', yerr=tabla['tiempoCD_y'], color=color_query)
ax.set_yscale('log')

# Añadir la línea horizontal en y=3600
# ax.axhline(y=3600, color='r', linestyle='--', label='3600 seconds (1 hour)')
ax.axhline(y=3600, color='r', linestyle='--')

# Etiquetas y formato
ax.set_xticks(x)
ax.set_xticklabels(tabla.index, ha='right')
ax.set_xlabel('Query')
ax.set_ylabel('Average time in seconds (logarithmic scale)')
#ax.set_title('Time without and with physical design')
ax.legend(loc='upper left' )#, bbox_to_anchor=(0.02, 0.95))

plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.savefig("Time10GB.jpg", bbox_inches='tight', dpi=300)
plt.show()
#files.download('Time30GB.jpg')
```



In [1875]:

```
import numpy as np
import matplotlib.pyplot as plt
```

```
x = np.arange(len(tabla))   # posición de cada grupo
bar_width = 0.4

# Crear gráfico
fig, ax = plt.subplots(figsize=(10, 5))

# Colores definidos
color_validate = '#1f77b4'  # azul
color_query = '#ff7f0e'     # naranja
color_query = '#FFD700'     # naranja

# Dibujar barras
ax.bar(x - bar_width/2, tabla['energySD_x'], width=bar_width, label='No design', yerr=ta
bla['energySD_y'], color=color_validate)
ax.bar(x + bar_width/2, tabla['energyCD_x'], width=bar_width, label='With physical desig
n', yerr=tabla['energyCD_y'], color=color_query)
ax.set_yscale('log')

# Etiquetas y formato
ax.set_xticks(x)
ax.set_xticklabels(tabla.index, ha='right')
ax.set_xlabel('Query')
ax.set_ylabel('Energy (joules)')
#ax.set_title('Energy consumption without and with physical design')
#ax.legend()
ax.legend(loc='upper left', bbox_to_anchor=(0.02, 0.95))

plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tight_layout()
plt.savefig("Energy10GB.jpg", bbox_inches='tight', dpi=300)
plt.show()
```



In [1876]:

```
def relativeSTD1(fila):
    return round((fila["tiempoSD_y"] /fila["tiempoSD_x"]),2)
def relativeSTD2(fila):
    return round((fila["tiempoCD_y"] /fila["tiempoCD_x"]),2)
def relativeSTD3(fila):
    return round((fila["energySD_y"] /fila["energySD_x"]),2)
def relativeSTDI4(fila):
    return round((fila["energyCD_y"] /fila["energyCD_x"]),2)
```

Medias: tiempoSD_x tiempoCD_x energySD_x energyCD_x Desviación estándar: tiempoSD_y tiempoCD_y energySD_y energyCD_y coeficiencientes de variación (de_relativa): RSEC_tSD RSEC_tCD RSEC_eSD RSEC_eCD

In [1877]:

```
tabla["RSEC_tSD"] = tabla.apply(relativeSTD1, axis=1)
tabla["RSEC_tCD"] = tabla.apply(relativeSTD2, axis=1)
tabla["RSEC_eSD"] = tabla.apply(relativeSTD3, axis=1)
tabla["RSEC_eCD"] = tabla.apply(relativeSTDI4, axis=1)
tabla
```

Out[1877]:

	tiempoSD_x	tiempoCD_x	energySD_x	energyCD_x	tiempoSD_y	tiempoCD_y	energySD_y	energyCD_y	RSEC_tSD	R
Query										
1	28.62	18.77	26.19	39.22	3.58	6.27	8.67	52.18	0.13	
2	2.39	0.79	3.30	1.44	0.32	0.19	2.03	1.55	0.13	
3	112.83	74.00	86.01	57.95	8.71	2.79	10.63	2.40	0.08	
4	1.82	0.15	2.18	0.59	0.27	0.05	0.90	0.42	0.15	
5	13.95	14.36	12.21	12.47	0.75	0.87	1.99	1.72	0.05	
6	2.16	0.41	2.43	0.62	0.31	0.10	0.94	0.44	0.14	
7	6174.36	95.45	3157.77	149.03	0.00	9.22	0.00	16.54	0.00	
8	2.79	0.80	3.71	1.16	0.57	0.08	2.94	0.47	0.20	
9	18.71	14.02	18.39	13.86	0.89	1.65	1.67	3.12	0.05	
10	4074.47	4634.88	3234.78	3238.68	0.00	0.00	0.00	0.00	0.00	
11	9.59	8.36	8.35	7.40	5.38	4.53	3.98	4.08	0.56	
12	36.19	1.75	46.03	2.13	3.79	1.11	5.94	1.17	0.10	
13	121.19	NaN	2.21	NaN	379.06	NaN	2.07	NaN	3.13	
14	8.39	7.51	8.80	7.54	1.17	0.38	2.70	1.46	0.14	
15	1.78	0.16	1.94	0.43	0.15	0.06	0.63	0.26	0.08	
16	32.34	30.24	29.69	28.31	2.29	5.03	1.92	4.75	0.07	
17	9.44	0.86	9.16	1.28	0.60	0.14	1.44	0.73	0.06	
18	10.72	6.76	10.59	6.77	0.34	0.47	1.58	2.11	0.03	
19	364.81	280.86	330.55	240.08	18.68	5.54	23.18	12.71	0.05	
20	19.63	8.44	30.52	23.82	0.61	0.48	8.83	4.78	0.03	
21	3.58	0.46	4.72	1.04	0.33	0.07	2.30	0.53	0.09	
22	11081.07	94.32	3378.77	235.17	0.00	5.91	0.00	44.60	0.00	
23	2.32	0.31	3.12	0.55	0.44	0.04	2.44	0.21	0.19	
24	42.24	34.89	46.07	34.68	2.97	2.55	10.12	4.42	0.07	
25	32.25	0.36	27.52	1.17	0.82	0.09	2.24	0.90	0.03	

In [1878]:

```
r1=[]
r2=[]
for i in range(1,26):
  query1=data[data["Query"] ==i].loc[:, ["tiempoSD","tiempoCD"]]
  query2=data[data["Query"] ==i].loc[:, ["energySD","energyCD"]]
  statistic, p_value = stats.wilcoxon(query1["tiempoSD"],query1["tiempoCD"])
  r1.append((statistic,'= '+str(np.round(p_value, 3))))
  statistic, p_value = stats.wilcoxon(query2["energySD"],query2["energyCD"])
  r2.append((statistic,'= '+str(np.round(p_value, 3))))
tabla["W_t"] = r1
tabla["W_e"] = r2
tabla
```

Out[1878]:

Query	tiempoSD_x	tiempoCD_x	energySD_x	energyCD_x	tiempoSD_y	tiempoCD_y	energySD_y	energyCD_y	RSEC_tSD	R
1	28.62	18.77	26.19	39.22	3.58	6.27	8.67	52.18	0.13	
2	2.39	0.79	3.30	1.44	0.32	0.19	2.03	1.55	0.13	
3	112.83	74.00	86.01	57.95	8.71	2.79	10.63	2.40	0.08	
4	1.82	0.15	2.18	0.59	0.27	0.05	0.90	0.42	0.15	
5	13.95	14.36	12.21	12.47	0.75	0.87	1.99	1.72	0.05	
6	2.16	0.41	2.43	0.62	0.31	0.10	0.94	0.44	0.14	
7	6174.36	95.45	3157.77	149.03	0.00	9.22	0.00	16.54	0.00	
8	2.79	0.80	3.71	1.16	0.57	0.08	2.94	0.47	0.20	
9	18.71	14.02	18.39	13.86	0.89	1.65	1.67	3.12	0.05	
10	4074.47	4634.88	3234.78	3238.68	0.00	0.00	0.00	0.00	0.00	
11	9.59	8.36	8.35	7.40	5.38	4.53	3.98	4.08	0.56	
12	36.19	1.75	46.03	2.13	3.79	1.11	5.94	1.17	0.10	
13	121.19	NaN	2.21	NaN	379.06	NaN	2.07	NaN	3.13	
14	8.39	7.51	8.80	7.54	1.17	0.38	2.70	1.46	0.14	
15	1.78	0.16	1.94	0.43	0.15	0.06	0.63	0.26	0.08	
16	32.34	30.24	29.69	28.31	2.29	5.03	1.92	4.75	0.07	
17	9.44	0.86	9.16	1.28	0.60	0.14	1.44	0.73	0.06	
18	10.72	6.76	10.59	6.77	0.34	0.47	1.58	2.11	0.03	
19	364.81	280.86	330.55	240.08	18.68	5.54	23.18	12.71	0.05	
20	19.63	8.44	30.52	23.82	0.61	0.48	8.83	4.78	0.03	

Query	tiempoSD_x	tiempoCD_x	energySD_x	energyCD_x	tiempoSD_y	tiempoCD_y	energySD_y	energyCD_y	RSEC_tSD	R
21	3.58	0.46	4.72	1.04	0.33	0.07	2.30	0.53	0.09	
22	11081.07	94.32	3378.77	235.17	0.00	5.91	0.00	44.60	0.00	
23	2.32	0.31	3.12	0.55	0.44	0.04	2.44	0.21	0.19	
24	42.24	34.89	46.07	34.68	2.97	2.55	10.12	4.42	0.07	
25	32.25	0.36	27.52	1.17	0.82	0.09	2.24	0.90	0.03	

In [1879]:

```
tabla.fillna('-')
```

Out[1879]:

Query	tiempoSD_x	tiempoCD_x	energySD_x	energyCD_x	tiempoSD_y	tiempoCD_y	energySD_y	energyCD_y	RSEC_tSD	R
1	28.62	18.77	26.19	39.22	3.58	6.27	8.67	52.18	0.13	
2	2.39	0.79	3.30	1.44	0.32	0.19	2.03	1.55	0.13	
3	112.83	74.0	86.01	57.95	8.71	2.79	10.63	2.4	0.08	
4	1.82	0.15	2.18	0.59	0.27	0.05	0.90	0.42	0.15	
5	13.95	14.36	12.21	12.47	0.75	0.87	1.99	1.72	0.05	
6	2.16	0.41	2.43	0.62	0.31	0.1	0.94	0.44	0.14	
7	6174.36	95.45	3157.77	149.03	0.00	9.22	0.00	16.54	0.00	
8	2.79	0.8	3.71	1.16	0.57	0.08	2.94	0.47	0.20	
9	18.71	14.02	18.39	13.86	0.89	1.65	1.67	3.12	0.05	
10	4074.47	4634.88	3234.78	3238.68	0.00	0.0	0.00	0.0	0.00	
11	9.59	8.36	8.35	7.4	5.38	4.53	3.98	4.08	0.56	
12	36.19	1.75	46.03	2.13	3.79	1.11	5.94	1.17	0.10	

Query	tiempoSD_x	tiempoCD_x	energySD_x	energyCD_x	tiempoSD_y	tiempoCD_y	energySD_y	energyCD_y	RSEC_tSD	R
13	121.19	-	2.21	-	379.06	-	2.07	-	3.13	
14	8.39	7.51	8.80	7.54	1.17	0.38	2.70	1.46	0.14	
15	1.78	0.16	1.94	0.43	0.15	0.06	0.63	0.26	0.08	
16	32.34	30.24	29.69	28.31	2.29	5.03	1.92	4.75	0.07	
17	9.44	0.86	9.16	1.28	0.60	0.14	1.44	0.73	0.06	
18	10.72	6.76	10.59	6.77	0.34	0.47	1.58	2.11	0.03	
19	364.81	280.86	330.55	240.08	18.68	5.54	23.18	12.71	0.05	
20	19.63	8.44	30.52	23.82	0.61	0.48	8.83	4.78	0.03	
21	3.58	0.46	4.72	1.04	0.33	0.07	2.30	0.53	0.09	
22	11081.07	94.32	3378.77	235.17	0.00	5.91	0.00	44.6	0.00	
23	2.32	0.31	3.12	0.55	0.44	0.04	2.44	0.21	0.19	
24	42.24	34.89	46.07	34.68	2.97	2.55	10.12	4.42	0.07	
25	32.25	0.36	27.52	1.17	0.82	0.09	2.24	0.9	0.03	

In [1880]:

```
def mean_std1(fila):
    return str(fila["tiempoSD_x"])+"±"+str(fila["tiempoSD_y"])
def mean_std2(fila):
    return str(fila["tiempoCD_x"])+"±"+str(fila["tiempoCD_y"])
def mean_std3(fila):
    return str(fila["energySD_x"])+"±"+str(fila["energySD_y"])
def mean_std4(fila):
    return str(fila["energyCD_x"])+"±"+str(fila["energyCD_y"])

tabla["Mean_tSD"] = tabla.apply(mean_std1, axis=1)
tabla["Mean_tCD"] = tabla.apply(mean_std2, axis=1)
tabla["Mean_eSD"] = tabla.apply(mean_std3, axis=1)
tabla["Mean_eCD"] = tabla.apply(mean_std4, axis=1)
```

In [1881]:

```
tabla.drop(columns=['tiempoSD_x','tiempoCD_x','energySD_x','energyCD_x','tiempoSD_y','ti
empoCD_y','energySD_y','energyCD_y'], inplace=True)
```

In [1882]:

```
tabla
```

Out[1882]:

Query	RSEC_tSD	RSEC_tCD	RSEC_eSD	RSEC_eCD	W_t	W_e	Mean_tSD	Mean_tCD	Mean_eSD	Mean_eCD
1	0.13	0.33	0.33	1.33	(0.0, = 0.002)	(19.0, = 0.432)	28.62±3.58	18.77±6.27	26.19±8.67	39.22±52.18
2	0.13	0.24	0.62	1.08	(0.0, = 0.002)	(0.0, = 0.002)	2.39±0.32	0.79±0.19	3.3±2.03	1.44±1.55
3	0.08	0.04	0.12	0.04	(0.0, = 0.002)	(0.0, = 0.002)	112.83±8.71	74.0±2.79	86.01±10.63	57.95±2.4
4	0.15	0.33	0.41	0.71	(0.0, = 0.002)	(0.0, = 0.002)	1.82±0.27	0.15±0.05	2.18±0.9	0.59±0.42
5	0.05	0.06	0.16	0.14	(12.0, = 0.131)	(21.0, = 0.557)	13.95±0.75	14.36±0.87	12.21±1.99	12.47±1.72
6	0.14	0.24	0.39	0.71	(0.0, = 0.002)	(0.0, = 0.002)	2.16±0.31	0.41±0.1	2.43±0.94	0.62±0.44
7	0.00	0.10	0.00	0.11	(0.0, = 0.002)	(0.0, = 0.002)	6174.36±0.0	95.45±9.22	3157.77±0.0	149.03±16.54
8	0.20	0.10	0.79	0.41	(0.0, = 0.002)	(0.0, = 0.002)	2.79±0.57	0.8±0.08	3.71±2.94	1.16±0.47
9	0.05	0.12	0.09	0.23	(0.0, = 0.002)	(2.0, = 0.006)	18.71±0.89	14.02±1.65	18.39±1.67	13.86±3.12
10	0.00	0.00	0.00	0.00	(0.0, = 0.002)	(0.0, = 0.002)	4074.47±0.0	4634.88±0.0	3234.78±0.0	3238.68±0.0
11	0.56	0.54	0.48	0.55	(0.0, = 0.002)	(0.0, = 0.002)	9.59±5.38	8.36±4.53	8.35±3.98	7.4±4.08
12	0.10	0.63	0.13	0.55	(0.0, = 0.002)	(0.0, = 0.002)	36.19±3.79	1.75±1.11	46.03±5.94	2.13±1.17
13	3.13	NaN	0.94	NaN	(nan, = nan)	(nan, = nan)	121.19±379.06	nan±nan	2.21±2.07	nan±nan
14	0.14	0.05	0.31	0.19	(3.0, = 0.01)	(6.0, = 0.027)	8.39±1.17	7.51±0.38	8.8±2.7	7.54±1.46
15	0.08	0.38	0.32	0.60	(0.0, = 0.002)	(0.0, = 0.002)	1.78±0.15	0.16±0.06	1.94±0.63	0.43±0.26
16	0.07	0.17	0.06	0.17	(9.0, = 0.064)	(14.0, = 0.193)	32.34±2.29	30.24±5.03	29.69±1.92	28.31±4.75
17	0.06	0.16	0.16	0.57	(0.0, = 0.002)	(0.0, = 0.002)	9.44±0.6	0.86±0.14	9.16±1.44	1.28±0.73
18	0.03	0.07	0.15	0.31	(0.0, = 0.002)	(0.0, = 0.002)	10.72±0.34	6.76±0.47	10.59±1.58	6.77±2.11
19	0.05	0.02	0.07	0.05	(0.0, = 0.002)	(0.0, = 0.002)	364.81±18.68	280.86±5.54	330.55±23.18	240.08±12.71

Query	RSEC_tSD	RSEC_tCD	RSEC_eSD	RSEC_eCD	W_t	W_e	Mean_tSD	Mean_tCD	Mean_eSD	Mean_eCD
20	0.03	0.06	0.29	0.20	(0.0, = 0.002)	(3.0, = 0.02)	19.63±0.61	8.44±0.48	30.52±8.83	23.82±4.78
21	0.09	0.15	0.49	0.51	(0.0, = 0.002)	(0.0, = 0.002)	3.58±0.33	0.46±0.07	4.72±2.3	1.04±0.53
22	0.00	0.06	0.00	0.19	(0.0, = 0.002)	(0.0, = 0.002)	11081.07±0.0	94.32±5.91	3378.77±0.0	235.17±44.6
23	0.19	0.13	0.78	0.38	(0.0, = 0.002)	(0.0, = 0.002)	2.32±0.44	0.31±0.04	3.12±2.44	0.55±0.21
24	0.07	0.07	0.22	0.13	(0.0, = 0.002)	(0.0, = 0.002)	42.24±2.97	34.89±2.55	46.07±10.12	34.68±4.42
25	0.03	0.25	0.08	0.77	(0.0, = 0.002)	(0.0, = 0.002)	32.25±0.82	0.36±0.09	27.52±2.24	1.17±0.9

In [1883]:

```
tabla.columns
```

Out[1883]:

```
Index(['RSEC_tSD', 'RSEC_tCD', 'RSEC_eSD', 'RSEC_eCD', 'W_t', 'W_e',
       'Mean_tSD', 'Mean_tCD', 'Mean_eSD', 'Mean_eCD'],
      dtype='object')
```

In [1884]:

```
tabla=tabla[['Mean_tSD', 'RSEC_tSD','Mean_tCD', 'RSEC_tCD','Mean_eSD',  'RSEC_eSD', 'Mea
n_eCD', 'RSEC_eCD', 'W_t', 'W_e',
       ] + [col for col in tabla.columns if col not in ['RSEC_tSD', 'RSEC_tCD', 'RSEC_eS
D', 'RSEC_eCD', 'W_t', 'W_e',
       'Mean_tSD', 'Mean_tCD', 'Mean_eSD', 'Mean_eCD']]]
tabla.fillna('-',inplace=True)
```

```
<ipython-input-1884-c4a14a622a8a>:4: FutureWarning: Setting an item of incompatible dtype
is deprecated and will raise an error in a future version of pandas. Value '-' has dtype
incompatible with float64, please explicitly cast to a compatible dtype first.
  tabla.fillna('-',inplace=True)
```

In [1885]:

```
tabla.replace('nan±nan','-', inplace=True)
tabla.replace('(nan, = nan','-', inplace=True)
tabla
```

Out[1885]:

Query	Mean_tSD	RSEC_tSD	Mean_tCD	RSEC_tCD	Mean_eSD	RSEC_eSD	Mean_eCD	RSEC_eCD	W_t	W_e
1	28.62±3.58	0.13	18.77±6.27	0.33	26.19±8.67	0.33	39.22±52.18	1.33	(0.0, = 0.002)	(19.0, = 0.432)
2	2.39±0.32	0.13	0.79±0.19	0.24	3.3±2.03	0.62	1.44±1.55	1.08	(0.0, = 0.002)	(0.0, = 0.002)
3	112.83±8.71	0.08	74.0±2.79	0.04	86.01±10.63	0.12	57.95±2.4	0.04	(0.0, = 0.002)	(0.0, = 0.002)
4	1.82±0.27	0.15	0.15±0.05	0.33	2.18±0.9	0.41	0.59±0.42	0.71	(0.0, = 0.002)	(0.0, = 0.002)

Query	Mean_tSD	RSEC_tSD	Mean_tCD	RSEC_tCD	Mean_eSD	RSEC_eSD	Mean_eCD	RSEC_eCD	W_t	W_e
5	13.93±0.73	0.05	14.36±0.87	0.06	12.21±1.99	0.16	12.47±1.72	0.14	(12.0, = 0.131)	(21.0, = 0.557)
6	2.16±0.31	0.14	0.41±0.1	0.24	2.43±0.94	0.39	0.62±0.44	0.71	(0.0, = 0.002)	(0.0, = 0.002)
7	6174.36±0.0	0.00	95.45±9.22	0.1	3157.77±0.0	0.00	149.03±16.54	0.11	(0.0, = 0.002)	(0.0, = 0.002)
8	2.79±0.57	0.20	0.8±0.08	0.1	3.71±2.94	0.79	1.16±0.47	0.41	(0.0, = 0.002)	(0.0, = 0.002)
9	18.71±0.89	0.05	14.02±1.65	0.12	18.39±1.67	0.09	13.86±3.12	0.23	(0.0, = 0.002)	(2.0, = 0.006)
10	4074.47±0.0	0.00	4634.88±0.0	0.0	3234.78±0.0	0.00	3238.68±0.0	0.0	(0.0, = 0.002)	(0.0, = 0.002)
11	9.59±5.38	0.56	8.36±4.53	0.54	8.35±3.98	0.48	7.4±4.08	0.55	(0.0, = 0.002)	(0.0, = 0.002)
12	36.19±3.79	0.10	1.75±1.11	0.63	46.03±5.94	0.13	2.13±1.17	0.55	(0.0, = 0.002)	(0.0, = 0.002)
13	121.19±379.06	3.13	-	-	2.21±2.07	0.94	-	-	(nan, = nan)	(nan, = nan)
14	8.39±1.17	0.14	7.51±0.38	0.05	8.8±2.7	0.31	7.54±1.46	0.19	(3.0, = 0.01)	(6.0, = 0.027)
15	1.78±0.15	0.08	0.16±0.06	0.38	1.94±0.63	0.32	0.43±0.26	0.6	(0.0, = 0.002)	(0.0, = 0.002)
16	32.34±2.29	0.07	30.24±5.03	0.17	29.69±1.92	0.06	28.31±4.75	0.17	(9.0, = 0.064)	(14.0, = 0.193)
17	9.44±0.6	0.06	0.86±0.14	0.16	9.16±1.44	0.16	1.28±0.73	0.57	(0.0, = 0.002)	(0.0, = 0.002)
18	10.72±0.34	0.03	6.76±0.47	0.07	10.59±1.58	0.15	6.77±2.11	0.31	(0.0, = 0.002)	(0.0, = 0.002)
19	364.81±18.68	0.05	280.86±5.54	0.02	330.55±23.18	0.07	240.08±12.71	0.05	(0.0, = 0.002)	(0.0, = 0.002)
20	19.63±0.61	0.03	8.44±0.48	0.06	30.52±8.83	0.29	23.82±4.78	0.2	(0.0, = 0.002)	(5.0, = 0.02)
21	3.58±0.33	0.09	0.46±0.07	0.15	4.72±2.3	0.49	1.04±0.53	0.51	(0.0, = 0.002)	(0.0, = 0.002)
22	11081.07±0.0	0.00	94.32±5.91	0.06	3378.77±0.0	0.00	235.17±44.6	0.19	(0.0, = 0.002)	(0.0, = 0.002)
23	2.32±0.44	0.19	0.31±0.04	0.13	3.12±2.44	0.78	0.55±0.21	0.38	(0.0, = 0.002)	(0.0, = 0.002)
24	42.24±2.97	0.07	34.89±2.55	0.07	46.07±10.12	0.22	34.68±4.42	0.13	(0.0, = 0.002)	(0.0, = 0.002)
25	32.25±0.82	0.03	0.36±0.09	0.25	27.52±2.24	0.08	1.17±0.9	0.77	(0.0, = 0.002)	(0.0, = 0.002)

In [1885]:

```

```

In [1886]:

```
tabla.to_csv('/content/drive/MyDrive/Colab Notebooks/wilcoxon10GB.csv')
```

In [1887]:

```
data.isnull().sum()
```

Out[1887]:

	0
Query	0
Iteración	0
tiempoSD	0
tiempoCD	10
energySD	0
energyCD	10

dtype: int64

La interpretación de la prueba de Wilcoxon, o prueba de rangos con signo de Wilcoxon, se enfoca en determinar si existe una diferencia significativa entre dos grupos relacionados, utilizando la mediana de las diferencias entre los rangos. Si el valor p es menor o igual al nivel de significancia (generalmente 0.05), se rechaza la hipótesis nula, indicando que hay una diferencia significativa entre los grupos. En un estudio sobre el impacto de un nuevo medicamento, se podrían comparar los resultados de las pruebas de los pacientes antes y después del tratamiento utilizando la prueba de Wilcoxon. Si el valor p es significativo, se podría concluir que el medicamento tuvo un efecto.

In [1888]:

```
r1=[]
r2=[]
tabla_sig=pd.DataFrame()

for i in range(1,26):
  query1=data[data["Query"] ==i].loc[:, ["tiempoSD","tiempoCD"]]
  query2=data[data["Query"] ==i].loc[:, ["energySD","energyCD"]]
  statistic, p_value = stats.wilcoxon(query1["tiempoSD"],query1["tiempoCD"])
#  r1.append('Sig' if p_value<=0.05 else 'NoSig')
  r1.append('Sig' if p_value<=0.05 else ('NoSig' if p_value>0.05 else None))
  statistic, p_value = stats.wilcoxon(query2["energySD"],query2["energyCD"])
#  r2.append('Sig' if p_value<=0.05 else 'NoSig')
  r2.append('Sig' if p_value<=0.05 else ('NoSig' if p_value>0.05 else None))

tabla_sig["W_t"] = r1
tabla_sig["W_e"] = r2
tabla_sig.index = tabla_sig.index + 1
tabla_sig
```

Out[1888]:

	W_t	W_e
1	Sig	NoSig
2	Sig	Sig
3	Sig	Sig
4	Sig	Sig

	W_t	W_e
5	NoSig	NoSig
6	Sig	Sig
7	Sig	Sig
8	Sig	Sig
9	Sig	Sig
10	Sig	Sig
11	Sig	Sig
12	Sig	Sig
13	None	None
14	Sig	Sig
15	Sig	Sig
16	NoSig	NoSig
17	Sig	Sig
18	Sig	Sig
19	Sig	Sig
20	Sig	Sig
21	Sig	Sig
22	Sig	Sig
23	Sig	Sig
24	Sig	Sig
25	Sig	Sig

In [1889]:

```
tabla.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 25 entries, 1 to 25
Data columns (total 10 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   Mean_tSD  25 non-null     object 
 1   RSEC_tSD  25 non-null     float64
 2   Mean_tCD  25 non-null     object 
 3   RSEC_tCD  25 non-null     object 
 4   Mean_eSD  25 non-null     object 
 5   RSEC_eSD  25 non-null     float64
 6   Mean_eCD  25 non-null     object 
 7   RSEC_eCD  25 non-null     object 
 8   W_t       25 non-null     object 
 9   W_e       25 non-null     object 
dtypes: float64(2), object(8)
memory usage: 2.1+ KB
```

In [1890]:

```
NS=sum((tabla_sig['W_t']=="NoSig") & (tabla_sig['W_e']=="NoSig"))
NS
```

Out[1890]:

```
2
```

In [1891]:

```
S=sum((tabla_sig['W_t']=="Sig") & (tabla_sig['W_e']=="Sig"))
S
```

Out[1891]:

21

In [1892]:

```
N=sum((tabla_sig['W_t'] == None) & (tabla_sig['W_e']  == None))
N
```

Out[1892]:

0

In [1893]:

```
S/(NS+S)*100
```

Out[1893]:

91.30434782608695

In [1894]:

```
14/(8+14)
```

Out[1894]:

0.6363636363636364